

Md. Samiul Alam

Portfolio: md-samiul-alam.github.io/portfolio/ | LinkedIn: [linkedin.com/in/md-samiul-alam](https://www.linkedin.com/in/md-samiul-alam)

Contact: +1 (807) 355 9935 | Email: samiulalam555@gmail.com

TECHNICAL SKILLS SUMMARY

Languages	JavaScript, TypeScript, Python, Java, PHP, C++
Front-end	React, Next, Angular, Tailwind CSS, Material UI, Bootstrap, JQuery, HTML5, CSS3
Back-end	Node.js (Express), Next, Python Django, Fast API, Laravel
Database	SQL: MS SQL Server, PostgreSQL, NO SQL: MongoDB.
Project Management	Jira, Trello Board
Testing Frameworks	Mocha, Jest, Jasmine
DevOps	CI/CD, Docker Container
Cloud Solutions	AWS EC2, S3
Version Control	Git
AI Dev Tools	Cursor, Claude Code, Kiln
Leetcode Profile	https://leetcode.com/u/samiul-alam/

WORK EXPERIENCE

• **Autodesk Research**

- *Software Development Engineer II [June 2025 - Present]*

- Develop a functional prototype of an intelligent user interface.
- Work with the team to understand high-level goals, directions, and intent of the project.
- Design and implement a system architecture that supports extensibility and future development.
- Build and maintain the front-end using modern web technologies with **React**, **Next.JS**, **Tailwind CSS**, ensuring a smooth and user-friendly interface.
- Implement back-end systems to support data storage, processing, and integration with the front-end.
- Enable asynchronous and parallel real-time collaboration, ensuring smooth, synchronized user experiences across multiple users working simultaneously.
- Work in an agile, iterative environment, adapting to changing requirements and frequent updates.
- Contribute to the development of documentation, code reviews, and best practices for the team.
- Troubleshoot and resolve any issues related to integration, performance, or functionality.

• **LAKEHEAD APPLIED GEOMATICS RESEARCH LABORATORY**

- *Research Assistant [January 2022 - April 2024]*

- Developed a GIS-integrated full-stack app using Angular and Django, leveraging AWS EC2 for cloud deployment.
- Designed and implemented REST APIs for real-time geospatial data analysis, integrating a custom ML model.
- Improved system resilience and security on AWS, following SOC-2-aligned authentication protocols.

• **ENOSIS SOLUTIONS**

- *Senior Software Engineer [January 2021 - September 2022]*

- Led sprint planning and mentoring sessions while delivering production-ready, cloud-native applications.
- Deployed scalable microservices on **AWS EC2** using **Docker**, configured **CI/CD pipelines** to reduce deployment time.
- Conducted **peer code reviews**, implemented **linting**, and enforced coding standards to improve software quality.
- Managed real-time system monitoring and implemented **logging** to optimize system performance.
- Presented project progress **reports**, proposed **solutions** to challenges, and fostered transparent communication with clients.
- Identified and prioritized tasks in the **software development lifecycle (SDLC)** while streamlining project tracking and team collaboration using **Jira**.
- Increased the productivity of junior developers through hands-on mentoring, knowledge sharing, workflow guidance, and skill development initiatives.

- *Software Engineer [October 2018 - December 2020]*

- Developed and maintained high-performance **Single Page Applications (SPAs)** using **Angular**, increasing user engagement and optimizing component architecture, state management, & testing practices.
- Improved user satisfaction by resolving cross-browser compatibility issues, enhancing **UI/UX** with **Angular Material** and **Bootstrap**, and ensuring accessibility across all devices.
- Designed the database schema and optimized complex queries using **MSSQL server**, reducing the query execution time.

- Streamlined feature development by creating reusable components, custom modules, directives, pipes, and services, ensuring **scalability** and **maintainability** of the codebase.
- **Deployed** secure and scalable web applications on **AWS EC2**, configuring DNS and SSL certificates and containerized environments with **Docker** for consistent deployment.
- Integrated Okta Authentication API for secure user authentication and authorization in web applications while configuring **OAuth 2.0** and **OpenID Connect (OIDC)** protocols to enable seamless Single Sign-On (SSO) and token-based authentication.
- Reduced the number of production bugs by 25% by performing extensive **unit** and **end-to-end tests** using **Mocha** and **Jasmine**, ensuring high code coverage and identifying and resolving potential issues early in the development cycle, resulting in more stable and reliable applications.
- Implemented **SOLID** principles in the design of a new microservices architecture, resulting in improvement in code maintainability and scalability.
- Optimized message delivery using Redis Pub/Sub, reducing server load for broadcasting updates and notifications.
- Reduced the average response time of the API by 15% by developing and optimizing efficient RESTful APIs using Node.js (Express), implementing techniques such as caching, asynchronous programming, and database query optimization.
- Developed and integrated secure payment systems using **Stripe**, enabling seamless online transactions, and created comprehensive API documentation with **Swagger** to improve team collaboration.
- Improved system performance by utilizing **MongoDB** within a **microservices** architecture, enhancing scalability, availability, and support for rapid data growth.
- Established **CI/CD** pipelines with **GitLab**, improving deployment time significantly.

PROJECTS

- **Open Kitchen:** Open Kitchen is a web-based platform developed by Powerhouse Dynamics and Enosis Solutions to enhance energy efficiency, food safety, and operational efficiency in the food service industry. The project aims to reduce energy costs, improve food safety compliance, and streamline operations through real-time data monitoring and analysis.

Project Demo: cutt.ly/le6yuChL

Tools used: Angular, Material UI, High-charts.js, Node JS (Express), MS SQL Server

Contributions: My key contributions included implementing interactive data visualizations using High-charts.js, creating scalable Angular components for the front-end, and developing robust back-end RESTful APIs with secure Okta authentication using Node.js and Express. Optimized complex SQL queries to resolve bottleneck issues in REST APIs.

- **V-Alert:** V-Alert is a mobile app for rapid emergency notifications, enabling organizations to quickly send critical information to individuals via unlimited messaging and channels, with features like geofencing and system integration.

Project Demo: cutt.ly/Pe6yuMtA

Tools used: Node JS (Express), MongoDB, CircleCI, Docker, Websocket.

Contributions: Contributed to the project by developing the backend API using Node.js, Express, and MongoDB, implementing CRUD operations for efficient database management. Ensured smooth and automated deployments by integrating CI/CD pipelines. Improved application consistency and scalability by containerizing microservices with Docker. Enhanced user experience by implementing WebSocket functionality, enabling real-time, low-latency messaging within the application.

ACHIEVEMENTS & AWARDS

<i>Champion</i>	Intra AUST Programming Contest Spring 2017.
<i>Champion</i>	Intra AUST Programming Contest Spring 2016.
<i>2nd Place</i>	Intra AUST Programming Contest Spring 2015.

CERTIFICATIONS

- Cert Prep: Scrum Master | cutt.ly/Oe6yyYVI
- Angular: Creating and Hosting a Full-Stack Site | cutt.ly/Ke6yrptt
- Building Modern Projects with React | cutt.ly/be6ytFMq

EDUCATION

- **Master of Environmental Studies (Thesis)** [January 2022 - May 2024]
Dept. of Geography and the Environment, Lakehead University, Thunder Bay, ON, Canada
- **B.Sc in Computer Science & Engineering** [September 2014 - December 2018]
Ahsanullah University of Science and Technology, Dhaka, Bangladesh